B1 Flows, Fluctuations and Complexity — Model Solutions, TT 2016

Question 1 — Potential flow, sphere, stagnation points

(a) Velocity potential and Laplace equation

Problem. Show that an irrotational, incompressible flow admits $\mathbf{u} = \nabla\phi$ with $\nabla^2\phi = 0$.

Irrotational: $\boldsymbol{\nabla} \times \mathbf{u} = 0$ on a simply connected domain. Hence

$$\mathbf{u} = \nabla\phi.$$

Incompressible: $\nabla \cdot \mathbf{u} = 0$. Substitute:

$$\nabla \cdot (\nabla\phi) = 0.$$

Therefore

$$\boxed{\nabla^2\phi = 0.}$$

(b) Stagnation-point form and streamlines

Problem. Justify $\phi = W(x^2 + y^2 - 2z^2)$ near a stagnation point and find the streamlines.

At a stagnation point $\mathbf{u} = 0$, so ϕ is locally an extremum and the leading non-trivial term is quadratic. The geometry is axisymmetric about $\hat{\mathbf{z}}$ and the surface is impenetrable at $z = 0$, so ϕ depends on $\rho^2 = x^2 + y^2$ and z^2 only:

$$\phi = a(x^2 + y^2) + b\,z^2.$$

Apply $\nabla^2\phi = 0$:

$$2a + 2a + 2b = 0 \;\Rightarrow\; b = -2a.$$

Set $a = W$:

$$\boxed{\phi(x, y, z) = W(x^2 + y^2 - 2z^2).}$$

Velocity field:

$$\mathbf{u} = \nabla\phi = (2Wx,\ 2Wy,\ -4Wz).$$

Streamlines: $\mathrm{d}\mathbf{r} \parallel \mathbf{u}$. In cylindrical coords ($\rho^2 = x^2 + y^2$):

$$\frac{\mathrm{d}\rho}{2W\rho} = \frac{\mathrm{d}z}{-4Wz}.$$

Separate:

$$\frac{\mathrm{d}\rho}{\rho} = -\frac{1}{2}\frac{\mathrm{d}z}{z}.$$

Integrate:

$$\ln\rho = -\tfrac{1}{2}\ln z + \text{const}.$$

Exponentiate:

$$\boxed{\rho^2\,z = \text{const}} \qquad \text{(streamlines are cubic curves } z \propto 1/\rho^2\text{)}.$$

(c) Sphere moving through inviscid fluid: ϕ and $\mathbf{u}$

Problem. Sphere of radius R at rest in the sphere frame; uniform flow $-\mathbf{v} = -v\hat{\mathbf{z}}$ at infinity. Find $\phi(r, \theta)$ and $\mathbf{u}(\mathbf{r})$.

In the sphere's rest frame the fluid streams past with $\mathbf{u} \to -v\hat{\mathbf{z}}$ as $r \to \infty$ and $u_r = 0$ at $r = R$. Use

$$\phi(r, \theta) = A_0 + \frac{B_0}{r} + \left(A_1 r + \frac{B_1}{r^2}\right)\cos\theta.$$

Far-field $-vz = -vr\cos\theta$:

$$A_1 = -v, \qquad B_0 = 0, \qquad A_0 = 0.$$

Impose $u_r|_{r=R} = 0$ where $u_r = \partial_r\phi$:

$$\partial_r\phi = \left(A_1 - \frac{2B_1}{r^3}\right)\cos\theta.$$

Set to zero at $r = R$:

$$A_1 - \frac{2B_1}{R^3} = 0 \;\Rightarrow\; B_1 = \tfrac{1}{2}A_1 R^3 = -\tfrac{1}{2}vR^3.$$

Therefore

$$\boxed{\phi(r, \theta) = -v\left(r + \frac{R^3}{2r^2}\right)\cos\theta.}$$

Velocity components $u_r = \partial_r\phi$, $u_\theta = r^{-1}\partial_\theta\phi$:

$$u_r = -v\left(1 - \frac{R^3}{r^3}\right)\cos\theta,$$

$$u_\theta = v\left(1 + \frac{R^3}{2r^3}\right)\sin\theta.$$

$$\boxed{\mathbf{u} = -v\left(1 - \frac{R^3}{r^3}\right)\cos\theta\,\hat{\mathbf{r}} + v\left(1 + \frac{R^3}{2r^3}\right)\sin\theta\,\hat{\boldsymbol{\theta}}.}$$

(d) Stagnation points and local form

Stagnation: $u_r = u_\theta = 0$. From $u_\theta = 0$ on the sphere requires $\sin\theta = 0$:

$$\theta = 0,\ \pi \quad \text{at} \quad r = R.$$

Two stagnation points: front $S_+ = (R, \pi)$ (where flow $-v\hat{\mathbf{z}}$ first hits sphere) and back $S_- = (R, 0)$.

Expand near front pole $\theta = \pi$. Set $\theta = \pi - \vartheta$, $r = R + \xi$ with $\vartheta, \xi/R \ll 1$. Use Cartesian-like coords $z = -r\cos\theta \approx R + \xi - \frac{1}{2}R\vartheta^2$, $\rho = r\sin\theta \approx R\vartheta$.

Expand u_r:

$$u_r = -v\left(1 - \frac{R^3}{r^3}\right)\cos\theta.$$

Substitute $r = R + \xi$, $\cos\theta = -\cos\vartheta \approx -1$:

$$u_r \approx -v(1 - (1 - 3\xi/R))\,(-1) = -3v\,\xi/R.$$

Expand u_θ:

$$u_\theta = v\left(1 + \frac{R^3}{2r^3}\right)\sin\theta.$$

Substitute $r = R$, $\sin\theta = \sin\vartheta \approx \vartheta$:

$$u_\theta \approx v \cdot \tfrac{3}{2}\vartheta.$$

Translate to local Cartesian $\tilde{z} = -\xi$ (outward from sphere along stream), $\tilde{\rho} = R\vartheta$:

$$u_{\tilde{z}} = -u_r = +3v\,\xi/R = -3v\,\tilde{z}/R,$$

$$u_{\tilde{\rho}} = u_\theta = \tfrac{3v}{2R}\,\tilde{\rho}.$$

This matches $\mathbf{u} = \nabla\phi$ with $\phi = W(\tilde{\rho}^2 - 2\tilde{z}^2)$:

$$W = \frac{3v}{4R}.$$

$$\boxed{\text{Local stagnation form recovered with } W = 3v/(4R).}$$

(e) Pressure on the sphere

On the sphere $r = R$: $u_r = 0$, $u_\theta = \tfrac{3}{2}v\sin\theta$:

$$u^2|_R = \tfrac{9}{4}v^2\sin^2\theta.$$

Far-field: $u^2 \to v^2$, $p \to p_\infty$. Bernoulli:

$$\frac{u^2}{2} + \frac{p}{\rho} = \frac{v^2}{2} + \frac{p_\infty}{\rho}.$$

Solve for p:

$$p = p_\infty + \tfrac{1}{2}\rho\big(v^2 - u^2\big)\,.$$

Substitute u^2:

$$\boxed{p(\theta) = p_\infty + \tfrac{1}{2}\rho v^2\big(1 - \tfrac{9}{4}\sin^2\theta\big)\,.}$$

Maximum at stagnation points ($\theta = 0, \pi$): $p = p_\infty + \frac{1}{2}\rho v^2$. Minimum at equator $\theta = \pi/2$: $p = p_\infty - \frac{5}{8}\rho v^2$.

Question 2 — Lubrication: viscous coating on a rotating cylinder

(a) Slow-flow equations

Problem. Write the equations of slow viscous flow with interpretation and validity.

Stokes equations:

$$\nabla p = \mu\nabla^2\mathbf{u} + \rho\mathbf{g}, \tag{1}$$
$$\nabla\cdot\mathbf{u} = 0. \tag{2}$$

Equation (1) balances pressure gradient against viscous diffusion of momentum and body force; (2) is mass conservation. Validity: inertia $\rho(\mathbf{u}\cdot\nabla)\mathbf{u}$ and unsteady $\rho\partial_t\mathbf{u}$ negligible compared to viscous stress, i.e. Reynolds number

$$\mathrm{Re} = \frac{UL}{\nu} \ll 1,$$

and the Strouhal-based unsteady Reynolds number $L^2/(\nu T) \ll 1$.

(b) Lubrication equation, BCs, velocity profile

Thin film: $h \ll R$, so locally treat the cylinder surface as flat with θ along, z normal (out of cylinder). Lubrication: $\partial_z \gg \partial_\theta$, so $\nabla^2 u \approx \partial_z^2 u$. Tangential gravity component is $g\cos\theta$ (the angle θ measured so that $\theta = 0$ is the side where the surface motion goes upward against gravity component in $\hat{\boldsymbol{\theta}}$ direction; here $g\cos\theta$ acts opposite to u). Stokes in θ:

$$0 = -\partial_\theta p/(R\rho) + \nu\partial_z^2 u - g\cos\theta.$$

Pressure gradient negligible (free surface, thin film):

$$\boxed{\nu\,\frac{\partial^2 u}{\partial z^2} = g\cos\theta.}$$

Boundary conditions:

$$u(\theta,0) = U \quad \text{(no slip on cylinder)},$$
$$\mu\,\partial_z u\big|_{z=h} = 0 \quad \text{(no shear at free surface)}.$$

Integrate twice:

$$\nu\,\partial_z u = g\cos\theta\cdot z + c_1(\theta).$$

Apply free-surface BC at $z = h$:

$$c_1 = -gh\cos\theta.$$

Integrate again:

$$\nu u = g\cos\theta\Big(\tfrac{z^2}{2} - hz\Big) + c_2(\theta).$$

Apply no-slip at $z = 0$: $c_2 = \nu U$.

$$\boxed{u(\theta,z) = U + \Big(\tfrac{z^2}{2} - zh\Big)\frac{g\cos\theta}{\nu}.}$$

(c) Volume flux and the cubic for H

Compute $Q = \int_0^h u\,\mathrm{d}z$:

$$Q = Uh + \frac{g\cos\theta}{\nu}\Big[\tfrac{z^3}{6} - \tfrac{z^2 h}{2}\Big]_0^h.$$

Evaluate bracket:

$$\tfrac{h^3}{6} - \tfrac{h^3}{2} = -\tfrac{h^3}{3}.$$

Therefore

$$Q(\theta) = Uh - \frac{gh^3\cos\theta}{3\nu}.$$

Substitute $h = QH/U$:

$$Q = U\cdot\frac{QH}{U} - \frac{g\cos\theta}{3\nu}\left(\frac{QH}{U}\right)^3.$$

Simplify:

$$Q = QH - \frac{gQ^3H^3\cos\theta}{3\nu U^3}.$$

Divide by Q:

$$1 = H - \frac{gQ^2}{3\nu U^3}\,H^3\cos\theta.$$

Identify $\alpha = gQ^2/(3\nu U^3)$:

$$1 = H - \alpha H^3 \cos\theta.$$

Divide by H^3:

$$\boxed{\frac{1}{H^2} - \frac{1}{H^3} = \alpha \cos\theta.}$$

Why Q is constant. Mass conservation in steady state: $\partial_\theta(\rho Q) = 0$ along the (closed) film. The film is incompressible and the cylinder is infinite (no axial variation), so $\partial_\theta Q = 0$, i.e. Q is independent of θ.

(d) Perturbative $h(\theta)$ for $\alpha \ll 1$

Expand $H = H_0 + \alpha H_1 + \cdots$ in

$$\frac{1}{H^2} - \frac{1}{H^3} = \alpha \cos\theta.$$

At $O(1)$:

$$\frac{1}{H_0^2} - \frac{1}{H_0^3} = 0 \;\Rightarrow\; H_0 = 1.$$

At $O(\alpha)$, differentiate $f(H) = H^{-2} - H^{-3}$:

$$f'(H_0) = -2H_0^{-3} + 3H_0^{-4} = -2 + 3 = 1.$$

So

$$H_1 \cdot 1 = \cos\theta \;\Rightarrow\; H = 1 + \alpha\cos\theta + O(\alpha^2).$$

Convert to $h = QH/U$:

$$\boxed{h(\theta) = \frac{Q}{U}[1 + \alpha\cos\theta] + O(\alpha^2).}$$

(e) Existence of solution and maximum film thickness

Define $f(H) = H^{-2} - H^{-3}$. Solutions exist for all $\theta \in [0, 2\pi]$ iff $\alpha\cos\theta$ lies within the range of f, i.e.

$$|\alpha| \le f_{\max}.$$

Find $f_{\max}$: $f'(H) = -2H^{-3} + 3H^{-4} = 0$:

$$H_* = 3/2.$$

Evaluate:

$$f(H_*) = (2/3)^2 - (2/3)^3 = \tfrac{4}{9} - \tfrac{8}{27} = \tfrac{12-8}{27} = \tfrac{4}{27}.$$

Hence

$$\boxed{\alpha \le \tfrac{4}{27}}$$

for a solution to exist for all θ.

At the threshold, the film thickness reaches its maximum where $H = H_* = 3/2$:

$$h_{\max} = \frac{Q}{U}H_* = \frac{3Q}{2U}.$$

$$\boxed{h_{\max} = \tfrac{3}{2}\,Q/U.}$$

Physical mechanism. The flux Q has two contributions: dragging by the moving wall ($+Uh$) and gravity-driven drainage ($-gh^3 \cos\theta/3\nu$) which is maximal where $\cos\theta = 1$ (on the side where gravity opposes the wall motion). For thicker films, drainage grows like h^3 while drag grows like h, so a thick film cannot sustain the required upward flux. At $\alpha = 4/27$ the two balance with $\partial Q/\partial h = 0$, the marginal point of carrying capacity; thicker films collapse (drip off).

Question 3 — Bifurcations and a 2D phase portrait

(a) Fixed points of $\dot{x} = rx - \sin x$ at $r = 0$

At $r = 0$:

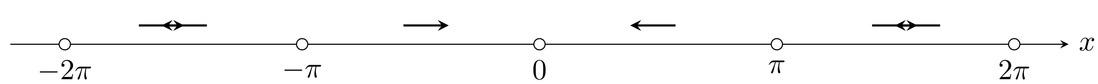
$$\dot{x} = -\sin x.$$

Fixed points: $\sin x = 0$, i.e. $x = n\pi$, $n \in \mathbb{Z}$.

Stability via $f'(x) = -\cos x$:

$$f'(n\pi) = -\cos(n\pi) = -(-1)^n.$$

Hence $x = 2k\pi$ are stable ($f' = -1$); $x = (2k+1)\pi$ are unstable ($f' = +1$).



Filled circles: stable; open circles: unstable. Flow points toward each $2k\pi$.

(b) $0 < r < 2$: bifurcations and counting

Fixed points satisfy $\sin x = rx$. Geometrically: intersections of $y = \sin x$ with the line $y = rx$.

Tangent bifurcations. At each saddle-node, $rx = \sin x$ and $r = \cos x$ tangentially. Eliminate:

$$\cos x_n \cdot x_n = \sin x_n \;\Rightarrow\; \tan x_n = x_n.$$

Solutions occur near $x_n \approx (n + \frac{1}{2})\pi$ for $n = 1, 2, \dots$ (positive side; symmetric on negative side). Bifurcation values:

$$r_n = \cos x_n.$$

For small r, the tangent bifurcations occur at large x_n where $\tan x \approx x$ gives $x_n \approx (n+\frac{1}{2})\pi - 1/[(n+\frac{1}{2})\pi]$. Thus

$$r_n = \cos x_n \approx (-1)^n \sin\left(\frac{1}{(n+\frac{1}{2})\pi}\right) \approx \frac{(-1)^n}{(n+\frac{1}{2})\pi}.$$

Take absolute values for $r > 0$:

$$\boxed{r_n \approx \frac{1}{(n+\frac{1}{2})\pi}, \quad n = 1, 2, \dots.}$$

Counting fixed points. For $0 < r \ll 1$, the line $y = rx$ intersects $\sin x$ near each maximum/minimum of $\sin x$ on $|x| \lesssim 1/r$. Number of crossings:

$$N(r) \sim \frac{2/r}{\pi} = \frac{2}{\pi r}.$$

$$\boxed{N(r) \approx 2/(\pi r) \text{ for } 0 < r \ll 1.}$$

At $r = 1$: line tangent to $\sin x$ at origin from above; the line touches $\sin x$ only at $x = 0$ (one fixed point). For $r > 1$ the line is steeper than $\sin x$ near 0, only one fixed point ($x = 0$, unstable since $f'(0) = r - 1 > 0$). At each $r = r_n$ a saddle-node pair annihilates as r increases, until at $r = 1$ all extra pairs are gone. For $r = 2$: still one fixed point ($x = 0$, unstable).

Stability: at $x = 0$, $f'(0) = r - 1$. So $x = 0$ is stable for $r < 1$, unstable for $r > 1$. (At $r = 1$ a transcritical-like change with the neighbouring pair occurs.) Pairs born in saddle-nodes alternate stable/unstable.

(c) Imaginary-axis dynamics

Substitute $x \mapsto iy$ ($y \in \mathbb{R}$):

$$i\dot{y} = r(iy) - \sin(iy) = iry - i\sinh y.$$

Divide by i:

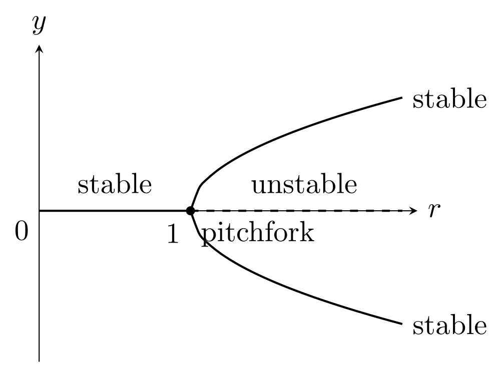
$$\dot{y} = ry - \sinh y.$$

Fixed points: $\sinh y = ry$.

For $r \le 1$: $\sinh y > ry$ for $y > 0$ (since $\sinh y \ge y \ge ry$ with equality only at $y = 0$), only $y = 0$ is a fixed point. Stability: $f'(0) = r - \cosh 0 = r - 1$. Stable for $r < 1$, unstable for $r > 1$.

For $r > 1$: $\sinh y$ grows faster than ry at large $|y|$, but at origin $\sinh y \approx y < ry$, so two new symmetric fixed points $\pm y_*(r)$ appear (pitchfork at $r = 1$).

Bifurcation diagram (supercritical pitchfork at $r = 1$):



(d) Phase portrait of the 2D system

Problem. $\dot{x} = y^3 - y$, $\dot{y} = x - y^2$.

Fixed points. Set $\dot{x} = \dot{y} = 0$:

$$y(y^2 - 1) = 0, \qquad x = y^2.$$

First gives $y \in \{0, \pm 1\}$.

$$y = 0 : \; x = 0.$$

$$y = \pm 1 : \; x = 1.$$

Three fixed points: $P_0 = (0,0)$, $P_+ = (1,1)$, $P_- = (1,-1)$.

Jacobian.

$$\mathsf{J} = \begin{pmatrix} 0 & 3y^2 - 1 \\ 1 & -2y \end{pmatrix}.$$

At $P_0 = (0,0)$:

$$\mathsf{J}_0 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Trace $\tau = 0$, det $\Delta = 1$. Eigenvalues $\pm i$: *centre* (linear). Nonlinear: try Hamiltonian / first integral. Note $\dot{x}/\dot{y} = (y^3 - y)/(x - y^2)$:

$$(x - y^2)\,\mathrm{d}x = (y^3 - y)\,\mathrm{d}y.$$

This is exact: $\mathrm{d}(\frac{x^2}{2} - xy^2) =?$ Test $\partial_y(x - y^2) = -2y$, $\partial_x(-(y^3 - y)) = 0$, not exact directly. Instead seek H with $\dot{x} = -\partial H/\partial y$, $\dot{y} = \partial H/\partial x$. Need $-\partial_y H = y^3 - y$ and $\partial_x H = x - y^2$:

$$H = \tfrac{x^2}{2} - xy^2 + \tfrac{y^4}{4} \cdot c?$$

Check $\partial_x H = x - y^2$ ✓. $\partial_y H = -2xy + g'(y)$; require $-(-2xy + g'(y)) = y^3 - y$:

$$2xy - g'(y) = y^3 - y.$$

But this depends on x, so the system is *not* Hamiltonian. The centre of the linearisation is therefore *not* guaranteed.

Compute divergence:

$$\nabla \cdot \mathbf{f} = \partial_x(y^3 - y) + \partial_y(x - y^2) = -2y.$$

Non-zero, so phase area is not conserved; near P_0 at $y = 0$ divergence vanishes to leading order. Linear analysis gives a centre; nonlinear terms decide. We treat P_0 as a (nonlinear) centre/focus; the phase portrait shows closed-curve-like trajectories near origin (sketched as a centre below).

At $P_\pm = (1, \pm 1)$:

$$\mathsf{J}_\pm = \begin{pmatrix} 0 & 2 \\ 1 & \mp 2 \end{pmatrix}.$$

P_+ $(y = 1)$:

$$\tau = -2, \quad \Delta = 0 \cdot (-2) - 2 \cdot 1 = -2.$$

$\Delta < 0$: *saddle*. Eigenvalues from $\lambda^2 + 2\lambda - 2 = 0$:

$$\lambda = -1 \pm \sqrt{3}.$$

Eigenvectors: for $\lambda_+ = -1 + \sqrt{3}$, $(\mathsf{J}_+ - \lambda_+ I)\mathbf{v} = 0$:

$$-\lambda_+ v_1 + 2v_2 = 0 \;\Rightarrow\; \mathbf{v}_+ \propto (2, \lambda_+) = (2, \sqrt{3} - 1).$$

For $\lambda_- = -1 - \sqrt{3}$: $\mathbf{v}_- \propto (2, -\sqrt{3} - 1)$.

P_- $(y = -1)$:

$$\tau = +2, \quad \Delta = 0 \cdot (2) - 2 \cdot 1 = -2.$$

$\Delta < 0$: *saddle*. Eigenvalues from $\lambda^2 - 2\lambda - 2 = 0$:

$$\lambda = 1 \pm \sqrt{3}.$$

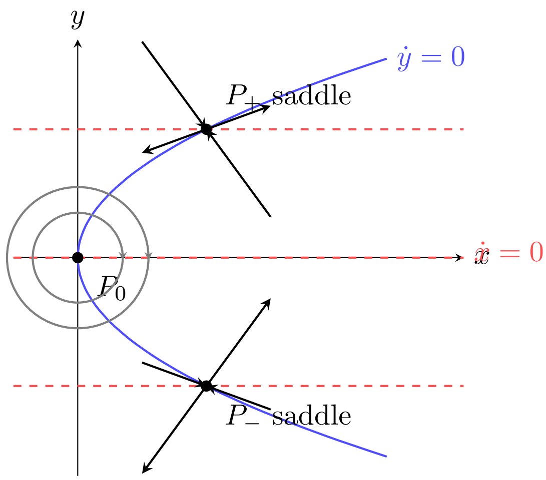
Nullclines.

$$\dot{x} = 0: \quad y = 0,\; y = \pm 1 \text{ (horizontal lines).}$$

$$\dot{y} = 0: \quad x = y^2 \text{ (parabola opening right).}$$

Saddles sit on parabola at $y = \pm 1$; centre at origin.

Phase portrait.



Summary:

$$\boxed{P_0 = (0,0) \text{ centre (linear)}, \quad P_\pm = (1, \pm 1) \text{ saddles.}}$$

Question 4 — Stochastic gene expression

(a) Deterministic solutions and steady states

Problem. Solve $\dot{R} = k_R - \gamma_R R$, $\dot{P} = k_P R - \gamma_P P$ for $R(t)$ and steady states.

Linear ODE for R:

$$\dot{R} + \gamma_R R = k_R.$$

General solution:

$$R(t) = \frac{k_R}{\gamma_R} + \left(R_0 - \frac{k_R}{\gamma_R} \right) e^{-\gamma_R t}.$$

$$\boxed{R(t) = \frac{k_R}{\gamma_R} \left(1 - e^{-\gamma_R t}\right) + R_0 e^{-\gamma_R t}.}$$

Steady state: $\dot{R} = 0$:

$$\langle R \rangle_{ss} = \frac{k_R}{\gamma_R}.$$

Then $\dot{P} = 0$:

$$\langle P \rangle_{ss} = \frac{k_P}{\gamma_P} \langle R \rangle_{ss} = \frac{k_P k_R}{\gamma_P \gamma_R}.$$

$$\boxed{\langle R \rangle_{ss} = k_R / \gamma_R, \qquad \langle P \rangle_{ss} = k_P k_R / (\gamma_P \gamma_R).}$$

Numerical estimate. Transcribed in 1 min: $k_R = 1$ molecule/min. RNA lifetime $1/\gamma_R = 10$ min, so $\gamma_R = 0.1 \text{ min}^{-1}$:

$$\langle R \rangle_{ss} = 1/0.1 = 10.$$

$$\boxed{\langle R \rangle_{ss} \approx 10 \text{ RNA molecules per cell.}}$$

(b) Mean-square fluctuations in R

Linearise about the steady state: $R = \langle R \rangle + \delta R$, drive by noise $\eta_R(t)$:

$$\delta \dot{R} = -\gamma_R \, \delta R + \eta_R(t).$$

Fourier transform $\delta R(t) = \int \frac{\mathrm{d}\omega}{2\pi} \, \delta \tilde{R}(\omega) e^{-i\omega t}$:

$$-i\omega \, \delta \tilde{R} = -\gamma_R \, \delta \tilde{R} + \tilde{\eta}_R.$$

Solve:

$$\delta \tilde{R}(\omega) = \frac{\tilde{\eta}_R(\omega)}{\gamma_R - i\omega}.$$

Power spectrum from $\langle \eta_R(t) \eta_R(t-t') \rangle = q_R \delta(t')$:

$$\langle \tilde{\eta}_R(\omega) \tilde{\eta}_R^*(\omega') \rangle = 2\pi q_R \, \delta(\omega - \omega').$$

Hence

$$\langle |\delta \tilde{R}(\omega)|^2 \rangle = \frac{q_R}{\gamma_R^2 + \omega^2} \cdot 2\pi \delta(0)_{\omega = \omega'}.$$

By Wiener–Khinchin:

$$\langle \delta R^2 \rangle = \int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi} \, \frac{q_R}{\gamma_R^2 + \omega^2}.$$

Apply standard integral with $a = \gamma_R$:

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}\omega}{2\pi(\omega^2 + \gamma_R^2)} = \frac{1}{2\gamma_R}.$$

Therefore

$$\boxed{\langle \delta R^2 \rangle = \frac{q_R}{2\gamma_R}.}$$

(c) Fano factor for P

Given

$$\langle \delta P^2 \rangle = \langle P \rangle \left(1 + \frac{k_P}{\gamma_R + \gamma_P} \right).$$

Fano factor:

$$F = \frac{\langle \delta P^2 \rangle}{\langle P \rangle} = 1 + \frac{k_P}{\gamma_R + \gamma_P}.$$

Manipulate: factor γ_R from denominator:

$$F = 1 + \frac{k_P/\gamma_R}{1 + \gamma_P/\gamma_R}.$$

Identify

$$b = \frac{k_P}{\gamma_R}, \qquad \phi = \frac{\gamma_P}{\gamma_R}.$$

$$\boxed{F = 1 + \frac{b}{1 + \phi}.}$$

Interpretation.

- $b = k_P/\gamma_R$: *burst size* — average number of protein molecules produced from a single mRNA before it is degraded (translation rate × mRNA lifetime).

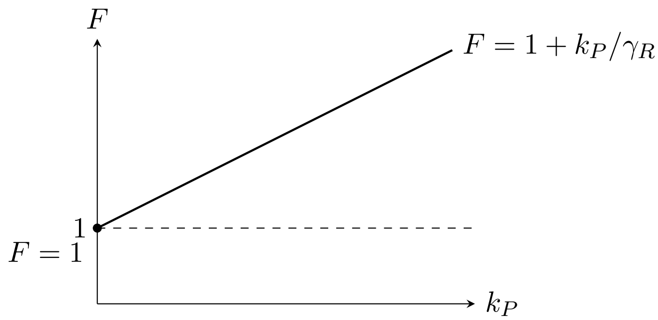
- $\phi = \gamma_P/\gamma_R$: ratio of protein-to-mRNA degradation rates. Since mRNA is much less stable than protein, $\gamma_R \gg \gamma_P$, so $\phi \ll 1$.

Poissonian limit. Pure Poisson ($b = 0$, no bursting): $F = 1$.

Dependence on k_P. With $\phi \ll 1$ ($1 + \phi \approx 1$),

$$F \approx 1 + b = 1 + k_P/\gamma_R.$$

F grows linearly in the translation rate k_P.



Larger k_P produces more proteins per mRNA burst, so the variance grows super-Poissonian; bursting is the source of the excess noise.

(d) Intrinsic vs. extrinsic noise

Intrinsic noise. Stochastic fluctuations originating from the inherent randomness of biochemical reactions (transcription, translation, degradation) within a single cell, given fixed cellular conditions. It is independent across two identical reporter genes in the same cell.

Extrinsic noise. Fluctuations arising from cell-to-cell differences in shared cellular factors (RNA polymerase, ribosome counts, cell volume, metabolic state). It correlates the expression of two reporter genes in the same cell.

Beneficial roles of intrinsic noise.

- *Phenotypic diversity / bet-hedging*: a clonal population samples multiple expression states, raising the chance that some cells survive sudden environmental change (e.g. bacterial persistence).

- *Stochastic differentiation*: in development or competence switching (e.g. *B. subtilis* competence), noise breaks symmetry between cells and triggers cell-fate decisions.

Detrimental roles.

- *Loss of fidelity*: noisy expression of essential housekeeping proteins compromises homeostasis.

- *Disease*: stochastic mis-regulation contributes to oncogenic states and developmental defects.

- *Reduced precision*: noise limits the accuracy of signalling, sensing, and gene-regulatory circuit responses.